2.4 隐函数及由参数方程所确定的函数的导数

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2. 已知 $y=f(x)$ 由方程 $\cos(xy)-\ln y+x=1$ ① 确定，则 $\lim\limits_{n\to\infty} n\left[f\left(\frac{2}{n}\right)-f(0)\right]=$ A ( )

A. 2　　B. 1　　C. -1　　D. -2

$x=0$

$-\sin(xy)\cdot(y+xy')-\frac{1}{y}\cdot y'+1=0$ ②

将 $x=0$ 代入 ① 式，$1-\ln y=1 \Rightarrow y=1$

将 $x=0$，$y=1$ 代入 ②，$-y'+1=0 \Rightarrow y'_{(0)}=1$

导数（或相关考点）+ 极限 ⇒ 导数定义.

$$\lim_{n\to\infty}\frac{f(\frac{2}{n})-f(0)}{\frac{1}{n}}$$

$$=\lim_{n\to\infty}2\frac{f(0+\frac{2}{n})-f(0)}{\frac{2}{n}}$$

$$=2f'_{+}(0)$$

$$\lim_{\Delta x\to 0}\frac{f(x_0+\boxed{\Delta x})-f(x_0)}{\boxed{\Delta x}}$$

$$=f'(x_0)$$

3. 已知 $y=x^{\frac{1}{x}}$，则 $\frac{dy}{dx}=$ B ( )

A. $\frac{1-\ln x}{x^2}$　　B. $x^{\frac{1}{x}-2}(1-\ln x)$　　C. $x^{\frac{1}{x}-1}(1-\ln x)$　　D. $x^{\frac{1}{x}-2}(\ln x-1)$

$$\ln y=\ln x^{\frac{1}{x}}=\frac{1}{x}\ln x$$

$$\frac{1}{y}y' = -\frac{1}{x^2}\cdot\ln x + \frac{1}{x}\cdot\frac{1}{x}$$

$$y' = x^{\frac{1}{x}}\cdot\left(\frac{1-\ln x}{x^2}\right)$$

$$= x^{\frac{1}{x}}\cdot x^{-2}\cdot(1-\ln x)$$

$$= x^{\frac{1}{x}-2}\cdot(1-\ln x)$$

5. 设$\begin{cases} x = \ln(1+t), \\ y = \frac{1}{1+t}, \end{cases}$ 则$\frac{d^2y}{dx^2} =$ (B)

A. 1　　B. $\frac{1}{1+t}$　　C. $\frac{1}{-1+t}$　　D. $\frac{-1}{1+t}$

$$\frac{dy}{dx} = y'_x = \frac{y'_t}{x'_t} \qquad \frac{d^2y}{dx^2} = \frac{(y'_x)'_t}{x'_t}$$

$$y'_t = -\frac{1}{(1+t)^2}$$

$$x'_t = \frac{1}{1+t}$$

$$\frac{dy}{dx} = \frac{y'_t}{x'_t} = \frac{-\frac{1}{(1+t)^2}}{\frac{1}{1+t}} = -\frac{1}{1+t}$$

$$\frac{d^2y}{dx^2} = \frac{+\frac{1}{(1+t)^2}}{\frac{1}{1+t}} = \frac{1}{1+t}$$

1. 设函数 $y = y(x)$ 由方程 $\ln(x^2+y) = x^3y + \sin x$ 所确定，则$\left.\frac{dy}{dx}\right|_{x=0} =$ _____.

①

$$\frac{1}{x^2+y}\cdot(2x+y') = 3x^2y + x^3\cdot y' + \cos x \quad ②$$

将 $x=0$ 代入 ①式

$$\ln y = 0 \Rightarrow y = 1$$

将 $x=0$, $y=1$ 代入 ②

$$y' = 1$$

$y' = 1$

4. 设 $y = y(x)$ 由 $\begin{cases} y = f(t), \\ x = f(e^{2t} - 1) \end{cases}$ 所确定，其中 f 可导，且 $f'(0) \neq 0$，则 $\left.\frac{dy}{dx}\right|_{t=0} =$ $\underline{\ 1/2\ }$.

$$\frac{dy}{dx} = \frac{y'_t}{x'_t}$$

$$y'_t = f'(t)$$

$$x'_t = f'(e^{2t}-1) \cdot 2e^{2t}$$

$$\left.\frac{dy}{dx}\right|_{t=0} = \frac{f'(0)}{f'(0) \cdot 2} = 1/2$$

6. 若由参数方程 $\begin{cases} x = \ln\cos t, \\ y = a\sec t \end{cases}$ 所确定的函数 $y = y(x)$ 满足 $\frac{dy}{dx} = y + e^{-x}$，则常数 $a =$ _____.

$$\frac{dy}{dx} = \frac{y'_t}{x'_t}$$

$$(\sec x)' = \sec x \cdot \tan x$$

$$y'_t = a\sec t \cdot \tan t$$

$$x'_t = \frac{1}{\cos t} \cdot (-\sin t) = -\tan t$$

$$\frac{dy}{dx} = \frac{a\sec t \cdot \tan t}{-\tan t} = -a\sec t$$

$$-a\sec t = a\sec t + e^{-\ln\cos t}$$

$$\frac{1}{e^{\ln\cos t}} = \frac{1}{\cos t}$$

$e^{\ln\cos t}$ $\cos t$

$$-a\sec t = a\sec t + \sec t$$

$e^{\ln f(x)} = f(x)$

$$\Rightarrow \quad -a = a+1$$

$$\Rightarrow \quad a = -\frac{1}{2}$$

1. 设函数 $y = y(x)$ 由方程 $\sin(x^2+y^2)+e^x-xy^2=0$ 所确定，求$\frac{dy}{dx}$.

$$\cos(x^2+y^2)\cdot(2x+2y\cdot y')+e^x-(y^2+x\cdot 2y\cdot y')=0$$

$$y'[2y\cdot\cos(x^2+y^2)-2xy] = -2x\cdot\cos(x^2+y^2)-e^x+y^2$$

$$y' = \frac{-2x\cdot\cos(x^2+y^2)-e^x+y^2}{2y\cdot\cos(x^2+y^2)-2xy}$$

3. 已知方程 $x^2+xy+y^2=4$ 所确定的隐函数为 $y=y(x)$，求$\frac{dy}{dx}$ 与$\frac{d^2y}{dx^2}$.

$$2x+y+xy'+2y\cdot y'=0 \quad ①$$

$$2+y'+y'+xy''+2(y'\cdot y'+y\cdot y'')=0 \quad ②$$

由①得 $y'(x+2y)=-2x-y$

$$y' = \frac{-2x-y}{x+2y} = -\frac{y+2x}{2y+x}$$

由②得

$$y''(x+2y) = -2-2y'-2y'^2$$

$$y'' = -\frac{y'+y'^2+1}{}$$

$$y'' = -2\,\frac{y'+y'^2+1}{x+2y}$$

$$= -\frac{24}{(x+2y)^3}$$

5. 设 $y = x^x + x^a + a^x + a^a (x > 0, a > 0$ 且 $a \neq 1)$，求 $\frac{dy}{dx}$.

x^x — 幂指，对数求导法.

$y_1 = x^x$，$\ln y_1 = x\ln x$

$\frac{1}{y_1}\cdot y_1' = \ln x + x\cdot\frac{1}{x} = \ln x + 1$

$y_1' = x^x\cdot(1+\ln x)$

x^a — 幂.

$(x^a)' = a x^{a-1}$

a^x — 指.

$(a^x)' = a^x\cdot\ln a$

a^a — 常数

$(a^a)' = 0$

6. 设 $y = (1+x)^{\frac{1}{\cos x}}$，求 y'.

$$\ln y = \ln (1+x)^{\frac{1}{\cos x}} = \frac{1}{\cos x}\cdot\ln(1+x)$$

$$\frac{1}{y}\cdot y' = -\frac{1}{(\cos x)^2}\cdot(-\sin x)\cdot \ln(1+x)$$

$$+\frac{1}{\cos x}\cdot\frac{1}{1+x}$$

$$y' = (1+x)^{\frac{1}{\cos x}}\cdot\left(\qquad\qquad\qquad\qquad\right)$$

7. 已知 $y=\frac{(x+1)\sqrt{x-1}}{(x+4)^2e^x}$，求 y'.

$$\ln y = \ln\frac{(x+1)\sqrt{x-1}}{(x+4)^2\cdot e^x}$$

$$= \ln(x+1) + \ln\sqrt{x-1}$$

$$- \ln(x+4)^2 - \ln e^x$$

$\ln ab = \ln a + \ln b$

$\ln\frac{b}{a} = \ln b - \ln a$

$\ln a^b = b\ln a$

$$\ln y = \ln(x+1) + \frac{1}{2}\ln(x-1) - 2\ln(x+4) - x$$

$$\frac{1}{y}\cdot y' = \underline{\frac{1}{x+1} + \frac{1}{2}\cdot\frac{1}{x-1} - 2\frac{1}{x+4} - 1}$$

$$y' = \frac{(x+1)\sqrt{x-1}}{(x+4)^2\cdot e^x}\left(\qquad\qquad\qquad\right)$$

10. 设 $y=y(x)$ 由 $\begin{cases} x = f'(t), \\ y = \underline{tf'(t)-f(t)} \end{cases}$ 所确定，$f''(t)$ 存在且 $f''(t)\neq 0$，求 $\frac{d^2y}{dx^2}$.

$$\frac{dy}{dx} = \frac{y'_t}{x'_t} \qquad \frac{d^2y}{dx^2} = \frac{(y'_x)'_t}{x'_t}$$

$$y'_t = f'(t) + tf''(t) - f'(t) = tf''(t)$$

$$x'_t = f''(t)$$

$$\frac{dy}{dx} = \frac{y'_t}{x'_t} = \frac{t f''(t)}{f''(t)} = t$$

$$\frac{d^2y}{dx^2} = \frac{(y'_x)'_t}{x'_t} = \frac{1}{f''(t)}$$